AI Log: ARGUS (AI in Finance Case Competition, Fall 2026)

Sanskriti Tiwari | Firm: Charles Schwab (equity research) | Subject analyzed: NVIDIA (NVDA), Q1 FY26 to Q2 FY27 releases + FY26 10-K | Prototype: Streamlit app, github.com/sans-2186/Argus

1. Tasks, tools and why

Core AI task	Tool	Why / role in the deliverable
Research the firm's subject: gather and cross-check filings (earnings releases, FY26 10-K)	AI-assisted web/SEC search with direct reading of the filings	Fast cross-document lookup; every figure was then checked against the filing (Section 3). Populated data/*.json and financial_metrics.csv.
Build a working prototype for analysts	AI coding assistant (Claude for the V2 changes: Research Lab, UI, status fix, tests)	Built the Streamlit app, the matcher, the deterministic verifier and the tests under a deadline. I specified requirements and reviewed the output.
Support only: script, polish	AI assistant	Demo script and wording; not counted as a core task.

What AI does NOT do at runtime: the app contains no live LLM, Gemini or embedding call. Matching management statements to metrics uses a local rule-based matcher (hand-built financial concept ontology + token overlap + segment-conflict penalty). All arithmetic and the 4-category classification run in deterministic Python.

2. Key prompts, inputs and outputs

Prompt 1 (filings): "Cross-check the Q4 FY26 and Q1 FY27 GAAP-to-Non-GAAP reconciliations; identify methodology changes and whether Q4 FY26 non-GAAP figures changed." **Output:** from Q1 FY27 NVIDIA stopped excluding stock-based comp (SBC); Q4 FY26 non-GAAP OpEx went from $5,102M to $6,666M. **Prompt 2 (narrative vs. numbers):** "Compare the Q2 FY27 'compute is revenue' commentary with Q2 FY27 revenue, cash flow and balance sheet trends." **Output:** revenue +17.90% QoQ but operating cash flow -52.18% and receivables +54.90%, flagged as a *research lead* (CMP-NVDA-003). **Prompt 3 (V2 build):** add a Research Lab tab that filters existing records, shows evidence, and exports a brief; correct the AI-status labels; no new dependencies; do not change data or benchmark labels.

3. Checks I made

• **Recalculated restatement:** Q4 FY26 OpEx SBC in the reconciliation footnote = $1,217M (R&D) + $347M (SG&A) = $1,564M; $5,102M + $1,564M = $6,666M, exact $0 variance.
• **Recalculated guidance vs. actual:** FY26 GAAP OpEx $23,076M vs. $16,405M = +40.66%; non-GAAP $16,694M vs. $11,716M = +42.49%; both above the 'high-30%' (39.9%) ceiling.
• **Automated tests:** 21 unit tests pass (6 original + 15 new covering the Research Lab filters, brief contents, zero-baseline math and missing-evidence wording). An earlier test caught a rounding label error (-52.175% shown as -52.17%; standardized to -52.18%).
• **Caught an AI overclaim:** on review, the app text and engine docstring said live Gemini embeddings were active whenever an API key existed. Reading argus_engine.py showed no API call exists. I corrected the UI and docstring to: "Local concept-aware matching active. No live Gemini inference is being performed in this workflow."
• **Edge cases:** a zero baseline in the custom-evidence form used to crash with ValueError; it now returns n/a. Empty evidence lists are displayed as "missing recorded evidence, not proof that none exists." A $-sign rendering bug (math-mode garbling of dollar amounts) was found in a screenshot and fixed.

4. Accepted, changed, rejected

Rejected: plain keyword matching as the link engine (on my 12-pair labeled set at the default thresholds it scored precision 33.3%, recall 12.5% vs. the local matcher's 100%/100%); and splicing old and restated non-GAAP OpEx series (would show a false +46% jump, so both series are stored). **Changed:** AI-status wording (above); divergences are phrased as research leads with contrary evidence and alternative explanations, never accusations. **Accepted with caution:** the benchmark result. It is a 12-pair set and I wrote the ontology while looking at the same filings, so 100% is a sanity check, not evidence of generalization; I did not edit the data or benchmark labels to improve scores.

5. Limits, privacy and bias

Public filings only (SEC EDGAR / NVIDIA IR); no confidential or Finance Lab raw data was put into AI tools. One company, six quarters. Results depend on my hand-built concept ontology, which can embed my own blind spots. Next step: evaluate an LLM/embedding matcher on a much larger labeled set, keeping the Python verification layer. Proposed success measures: analyst review time per earnings release and the share of flagged leads analysts judge useful (not measured here).